

POWER BI, TABLEAU & MACHINE LEARNING


pandas


Power BI


+a|b|eau


TensorFlow


Machine Learning

 **seaborn**



matplotlib

 **python™**



A Brief About **UPFLAIRS**

UpFlairs is an innovative **educational technology** company with a clear mission to elevate the skills and employability of students throughout India.

Our dedicated team is committed to fostering the next generation of tech talent, equipping them with cutting-edge skills in emerging technologies and has educated more than 47K students all over the globe including IITs, NITs, Deemed Universities and other colleges.

We offer the courses that are most trending technologies of the recent era in the fields like **AI/ML, Data Science, Cloud Computing, DevOps, Full Stack Web Development, Embedded Systems, IoT, and Robotics**. These courses are meticulously designed to provide students with the practical skills and knowledge required to excel in tech-driven careers, making them not just job-ready but industry leaders of tomorrow.

We are not only limited to training the youth of the country but also provide Lab setups to various Colleges and Universities and Project solutions to other companies for the domains like **AI-ML, IOT , ROBOTICS AND CLOUD**.



POWER BI , TABLEAU & MACHINE LEARNING

Duration – 75 HOURS

MODULE 1 – 25 Hours (POWER BI)

1	Introduction to Power BI	- Overview of Business Intelligence & Power BI - Installing & Setting up Power BI Desktop - Connecting to Different Data Sources (Excel, SQL, Web, APIs) - Data Cleaning & Transformation using Power Query
	Mini Project	<i>Project 1: “Load and Clean Sales Data from E-Commerce Store”</i>
2	Data Modeling & DAX	- Understanding Data Relationships & Cardinality - Introduction to DAX (Data Analysis Expressions) - Common DAX Functions: SUM, AVERAGE, COUNT, FILTER, CALCULATE - Creating Calculated Columns, Measures, and Tables
	Mini Project	<i>Project 2: “Create Financial KPIs for Revenue Forecasting”</i>
3	Data Visualization & Report	- Building Interactive Dashboards - Types of Visuals (Bar, Pie, Map, KPI, Card, Table, Matrix) - Using Filters, Slicers, and Drill-Throughs - Publishing & Sharing Reports in Power BI Service
	Mini Project	<i>Project 3: “E-Commerce Sales Dashboard”</i>

MODULE 2 – 25 HOURS (TABLEAU)

4	Getting Started with Tableau	- Introduction to Tableau & Installation - Connecting to Data Sources (Excel, SQL, APIs) - Data Preparation & Data Types - Joins, Blends, and Relationships in Tableau
	Mini Project	<i>Project 4: “Connect to banking customer dataset and Clean data”</i>
5	Creating Charts & Visualizations	- Bar, Line, Scatter, and Pie Charts - Heatmaps, TreeMaps, and Bullet Charts - Dual-Axis & Combination Charts - Formatting & Best Practices for Dashboard Design
	Mini Project	<i>Project 5: “Create a Financial Dashboard for banking transactions”</i>
6	Advanced Tableau & Dashboard Creation	- Filters, Parameters & Sets - Creating Dashboards with Actions - Calculated Fields and Table Calculations - Storytelling with Data
	Mini Project	<i>Project 6: “Bank Loan Risk Analysis Dashboard”</i>

MODULE 3 - 25 HOURS (MACHINE LEARNING)

7	Introduction to Machine Learning	Supervised vs Unsupervised Learning
---	----------------------------------	---

	& Data Pre-processing	- Machine Learning Workflow & Pipeline - Python Libraries: NumPy, Pandas, Matplotlib, Scikit-Learn - Data Preprocessing: Handling Missing Values, Encoding, Feature Scaling - Hands-on: Load and preprocess customer behaviour data
	Mini Project	<i>Project 7: "Load and preprocess customer behavior data"</i>
8	Regression & Classification Models	- Linear Regression & Polynomial Regression - Logistic Regression for Classification Problems - Model Evaluation Metrics (MSE, R2 Score, Accuracy, Precision, Recall)
	Mini Project	<i>Project 8: "Predicting Real Estate Prices"</i>
9	Advanced ML Algorithm & Model Deployment	- Decision Trees & Random Forest - Support Vector Machines (SVM) - Clustering (K-Means, DBSCAN, Hierarchical) - Dimensionality Reduction (PCA) - Model Deployment using Flask/Streamlit
	Mini Project	<i>Project 8: "Customer Churn Prediction for a Telecom Company"</i>
	Capstone Project	<i>Capstone Project 1: "Retail Demand Forecasting & Insights Dashboard"</i>

NIT,
Jalandhar

Glance Upflairs

Delhi Technical
University

NIFFT,
Ranchi

Banasthali
Vidyapith

JECRC,
Jaipur

Arya College,
Jaipur

Poornima
College, Jaipur

SKIT, Jaipur



<https://upflairs.com>



9251494002, 6350417917



info@upflairs.com



100/110ps, Rana Sanga Marg, Near Kumbha Marg, Pratap Nagar, Jaipur , Rajasthan